

<i>typevar</i> , X	type variable	
<i>termvar</i> , x	term variable	
<i>label</i> , l, k	field label	
<i>index</i> , i, j, n, m	indices	
T, S, U	$::=$	type
	X	type variable
	$\{l_1 : T_1, \dots, l_n : T_n\}$	record
t	$::=$	term
	x	variable
	$\{\overline{l_i = t_i}^{i \in 1..n}\}$	record — complu with terminal
	$t.l$	projection
Γ, Δ	$::=$	type environment
	empty	
	$\Gamma, X <: T$	
	$\Gamma, x : T$	
	Γ, Γ'	M
	$\Gamma_1, \dots, \Gamma_n$	M
<i>formula</i>	$::=$	
	<i>judgement</i>	
	<i>formula</i> ₁ .. <i>formula</i> _n	
<i>terminals</i>	$::=$	
	$\vdash$	
	$<:$	
<i>Jtype</i>	$::=$	
	$\Gamma \vdash t : T$	term t has type T
<i>judgement</i>	$::=$	
	<i>Jtype</i>	
<i>user_syntax</i>	$::=$	
	<i>typevar</i>	
	<i>termvar</i>	
	<i>label</i>	
	<i>index</i>	
	T	
	t	
	Γ	
	<i>formula</i>	
	<i>terminals</i>	

$\boxed{\Gamma \vdash t : T}$ term t has type T

$$\begin{array}{c}
\frac{\Gamma \vdash t_1 : T_1 \quad \dots \quad \Gamma \vdash t_n : T_n}{\Gamma \vdash \{l_1 = t_1, \dots, l_n = t_n\} : \{l_1 : T_1, \dots, l_n : T_n\}} \quad \text{TY_RCD_DOTFORM} \\
\frac{\Gamma \vdash t : \{l_1 : T_1, \dots, l_n : T_n\}}{\Gamma \vdash t.l_j : T_j} \quad \text{TY_PROJ_DOTFORM}
\end{array}$$

$$\begin{array}{c}
\frac{\overline{\Gamma \vdash t_i : T_i}^i}{\Gamma \vdash \{\overline{l_i = t_i}^i\} : \{\overline{l_i : T_i}^i\}} \quad \text{TY_RCD_COMP} \\
\\
\frac{\Gamma \vdash t : \{\overline{l_i : T_i}^i\}}{\Gamma \vdash t.l_j : T_j} \quad \text{TY_PROJ_COMP} \\
\\
\frac{\overline{\Gamma \vdash t_i : T_i}^{i < n}}{\Gamma \vdash \{\overline{l_i = t_i}^{i < n}\} : \{\overline{l_i : T_i}^{i < n}\}} \quad \text{TY_RCD_COMP_U} \\
\\
\frac{\Gamma \vdash t : \{\overline{l_i : T_i}^{i < n}\}}{\Gamma \vdash t.l_j : T_j} \quad \text{TY_PROJ_COMP_U} \\
\\
\frac{\overline{\Gamma \vdash t_i : T_i}^{i \in 1..n}}{\Gamma \vdash \{\overline{l_i = t_i}^{i \in 1..n}\} : \{\overline{l_i : T_i}^{i \in 1..n}\}} \quad \text{TY_RCD_COMP_LU} \\
\\
\frac{\Gamma \vdash t : \{\overline{l_i : T_i}^{i \in 1..n}\}}{\Gamma \vdash t.l_j : T_j} \quad \text{TY_PROJ_COMP_LU} \\
\\
\frac{\Gamma \vdash t : \{l_0 : T_0, \dots, l_{n-1} : T_{n-1}\}}{\Gamma \vdash t.l_j : T_j} \quad \text{TY_PROJ_DOTFORM_MINUS} \\
\\
\frac{\Gamma \vdash t : \{\overline{l_i : T_i}^{i \in 0..n-1}\}}{\Gamma \vdash t.l_j : T_j} \quad \text{TY_PROJ_COMP_LU_MINUS}
\end{array}$$

Definition rules: 10 good 0 bad

Definition rule clauses: 20 good 0 bad